

PARKING MANAGEMENT SYSTEM

Vehicle Entry and Exit

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PROBLEM STATEMENT: Parking – Vehicle Entry & Exit

Features:

- Vehicle Details
- Parking Slot
- Vehicle Type
- Entry / Exit
- Fee Calculation

Algorithm

Topic: Parking – Vehicle Entry and Exit

Features: Vehicle details, parking slot, vehicle type, entry and exit, and fee calculation.

Requirements

1. Enter vehicle details such as Vehicle ID, Vehicle Number, and Owner Name.
2. Select the vehicle type such as Two-Wheeler, Car, or Other.
3. Check parking-slot availability and assign an available slot.
4. Record the vehicle entry date and time.
5. Store the vehicle, vehicle type, slot, and entry details.
6. When the vehicle exits, search and retrieve its parking details.
7. Record the exit date and time.
8. Calculate the total parking duration.
9. Calculate the parking fee according to vehicle type and parking duration.
10. Display the total fee and record payment.
11. Release the parking slot and mark it as available.
12. Update the exit details in the database.

Algorithm Steps

13. Start.
14. Enter vehicle details.
15. Select vehicle type.
16. Check whether a parking slot is available.
17. If no slot is available, display “Parking Slot Not Available” and stop.
18. If a slot is available, assign the slot to the vehicle.
19. Record the entry date and time.
20. Store the parking and vehicle details.
21. When the vehicle exits, search for the vehicle details.
22. Record the exit date and time.
23. Calculate the parking duration.
24. Calculate the parking fee based on vehicle type and duration.
25. Display the fee and record payment.
26. Release the parking slot.
27. Update the exit details.
28. Stop.

ER Diagram

Create a neat, simple, and easy-to-understand ER (Entity-Relationship) Diagram for a Parking Management System – Vehicle Entry and Exit, suitable for a 1st-semester BCA student.

Main Entities and Attributes

1. VEHICLE

- Vehicle_ID (Primary Key)
- Vehicle_Number
- Owner_Name
- Vehicle_Type

2. PARKING_SLOT

- Slot_ID (Primary Key)
- Slot_Number
- Slot_Type
- Slot_Status

3. PARKING_RECORD

- Record_ID (Primary Key)
- Entry_Time
- Exit_Time
- Parking_Duration

4. FEE

- Fee_ID (Primary Key)
- Fee_Amount
- Payment_Status

Relationships

- VEHICLE → parks in → PARKING_SLOT
- VEHICLE → has → PARKING_RECORD
- PARKING_SLOT → is assigned to → PARKING_RECORD
- PARKING_RECORD → generates → FEE

Requirements

- Show Primary Keys (PK) clearly.
- Use rectangles for entities.
- Use ovals for attributes.
- Underline primary key attributes.

- Use diamonds for relationships.
- Show cardinality such as 1:1 or 1:M where appropriate.
- Clearly represent the vehicle entry and exit process.
- Include vehicle details, parking slot, vehicle type, entry/exit time, parking duration, and fee calculation.
- Keep the diagram clean, properly aligned, and easy to understand.
- Do not add unnecessary entities or attributes.
- Use clear arrows/connecting lines between entities and relationships.
- Make it suitable for a college DBMS practical record.

Flowchart

Create a professional and simple flowchart for a DBMS project titled “Parking – Vehicle Entry and Exit”.

The flowchart should be suitable for a 1st-semester BCA practical/project record and should clearly show the complete process from vehicle entry to exit and fee calculation.

Features/Requirements

- Vehicle Details – Enter Vehicle ID, Vehicle Number, and Owner Name.
- Vehicle Type – Select vehicle type such as Two-Wheeler, Car, or Other.
- Parking Slot – Check slot availability and assign an available Parking Slot ID/Number.
- Entry – Record the vehicle’s entry date and time.
- Parking – Store vehicle, vehicle type, slot, and entry details in the system.
- Exit – Search the vehicle details when the vehicle leaves and record the exit date and time.
- Fee Calculation – Calculate the parking duration and determine the parking fee according to the vehicle type and duration.
- Payment – Display the calculated fee and record payment.
- Slot Release – Mark the parking slot as available after the vehicle exits.

Flowchart Sequence

Start – Enter Vehicle Details – Select Vehicle Type – Check Parking Slot Availability – Is Slot Available? – Assign Slot – Record Entry Time – Park Vehicle – Vehicle Exit Request – Retrieve Vehicle Details – Record Exit Time – Calculate Parking Duration – Calculate Fee – Display Fee – Payment – Release Parking Slot – Update Database – End

Use Standard Flowchart Symbols

- Oval: Start/End
- Parallelogram: Input/Output
- Rectangle: Process
- Diamond: Decision
- Arrows: Flow direction

Make the flowchart neat, logically connected, easy to understand, properly labelled, and vertically arranged. Use clear text and avoid unnecessary complexity.